9 GEODETIC AND CONTROL SURVEYS

NORTH AMERICAN DATUMS

1. Which of the following statements correctly describes the origin of the North American Datum (NAD) of 1927?

(A) The origin of NAD 27 is at Meades Ranch in Kansas.
(B) The origin of NAD 27 is geocentric.
(C) The geoidal height of the origin of NAD 27 is assumed to be zero.
(D) Both A and C are true.

Solution: NAD 27 is based on ellipsoidal parameters defined as the Clarke Spheroid of 1866.

A readjustment of the national network done to introduce Laplace azimuths yielded the geodetic positions now available on NAD 27. The origin of that readjustment is designated as

$$\phi = 39°13'26.''686\text{N}$$
$$\lambda = 98°32'30.''506\text{E}$$

The origin is located at Meades Ranch in Kansas, and the azimuth from Meades Ranch to Waldo is defined as $75°28'09.''64$. The geoidal height is assumed to be zero, and NAD 27 is not geocentric.

Answer is (D)

2. Which of the following statements correctly describes the North American Datum of 1983?

(A) The origin of NAD 83 is geocentric.
(B) NAD 83 is the horizontal control datum used throughout the western hemisphere.
(C) Coordinates based on NAD 83 have replaced those based on NAD 27 throughout the U.S.
(D) Both A and B are true.

Solution: NAD 83 is based on a geocentric origin. 250,000 points, including 600 stations whose positions had been determined by satellite geodesy, were used to constrain the datum to its geocentric origin.

NAD 83 has not completely replaced NAD 27 in the U.S. and is not used throughout the western hemisphere. NAD 83 is used in the U.S., Canada, Mexico, and Central America.

Answer is (A)

3. NAD 83 uses a figure to represent the earth that closely approximates its shape, but may be described mathematically by a few definite parameters. What is the shape?

(A) a biaxial ellipsoid rotated about its minor axis
(B) a mean earth ellipsoid
(C) a biaxial ellipsoid rotated about its major axis
(D) a triaxial ellipsoid

Solution: The shape of the earth is represented in NAD 83 by an ellipsoid of revolution that is biaxial and rotated about its minor axis. The actual shape of the earth is oblate—that is, flattened at the poles. Rotation approximates the shape.

Answer is (A)

4. Which of the following statements correctly characterizes the transformation of coordinates from NAD 27 to NAD 83?

(A) There is not one simple, consistent method to reliably transform coordinates from NAD 27 to NAD 83.

(B) The transformation of ellipsoidal coordinates in terms of $\Delta\phi$, $\Delta\lambda$, and Δh from NAD 27 to NAD 83 can be accurately accomplished with a simple, one-step mathematical formula, but the transformation of Cartesian coordinates cannot.

(C) The transformation of Cartesian coordinates in terms of Δx, Δy, and Δz from NAD 27 to NAD 83 can be accurately accomplished with a simple, one-step mathematical formula, but transformation of ellipsoidal coordinates cannot.

(D) The transformation of state plane coordinates expressed in NAD 27 to NAD 83 involves only a change in the origin of the coordinates.

Solution: Local distortions caused by low-quality original observations in NAD 27 coordinates are one source of the difficulty involved in transforming coordinates from that datum to NAD 83. Approximate methods of transformation can be designed by computing local translation components based on stations with three-dimensional coordinates common to both systems, but there is no universal one-step mathematical approach available.

The most reliable transformations are based on a return to the original observations and the performance of a readjustment of the entire region.

Answer is (A)

5. Flattening and the length of the semimajor axis are two parameters often used to define the reference ellipsoids associated with geodetic datums. What are the dimensions used to define the reference ellipsoids associated with NAD 27 and NAD 83?

	Datum	Reference Ellipsoid	Semimajor	Flattening
(A)	NAD 83	GRS 80	6,378,137.0 m	$\frac{1}{298.257222101}$
	NAD 27	Clarke 1866	6,378,206.4 m	$\frac{1}{294.9786982}$
(B)	NAD 83	Fischer 1960	6,378,166.0 m	$\frac{1}{298.36714202}$
	NAD 27	Clarke 1858	6,378,361.4 m	$\frac{1}{294.26982312}$
(C)	NAD 83	WGS 84	6,378,137.0 m	$\frac{1}{298.257223563}$
	NAD 27	Clarke 1866	6,378,206.4 m	$\frac{1}{294.9786982}$
(D)	NAD 83	WGS 72	6,378,135.0 m	$\frac{1}{298.2623209}$
	NAD 27	Clarke 1858	6,378,206.4 m	$\frac{1}{294.9786982}$

Solution: Clarke 1866 is the reference ellipsoid for NAD 27. GRS 80 is the reference ellipsoid for NAD 83. The WGS 84 ellipsoid is virtually identical to the GRS 80, but there is a slight difference in the flattening, probably due to the lack of sufficient significant figures in its original computation.

Answer is (A)

6. What is the geoid?

(A) The geoid is a surface defined by the earth's gravity.
(B) The geoid is a surface that is flattened without tearing or stretching.
(C) The geoid is a surface that would closely resemble the oceans' surfaces if they were completely still.
(D) Both A and C are true.

Solution: One way to understand the geoid is to imagine that trenches have been cut through the land masses of the earth, and water has been allowed to flow into them. The surface assumed by the water would then approximate the surface known as the *geoid*. The geoid is often called an *equipotential surface*. In other words, the direction of gravity is always perpendicular to the geoid.

Answer is (D)

7. How is the flattening, f, of a reference ellipsoid defined in terms of the semimajor axis, a, and the semiminor axis, b?

(A) $f = \dfrac{b-a}{b}$
(B) $f = \dfrac{a-b}{a}$
(C) $f = b - a$
(D) $f = \dfrac{a+b}{b}$

Solution: The flattening associated with a reference ellipsoid is conventionally expressed as

$$f = \frac{a-b}{a}$$

Answer is (B)

8. What is the cause of geoidal undulation?

(A) The presence of magnetically attractive ores within the earth cause th geoid to undulate.
(B) The center of mass of the earth does not coincide with the center of th geoid.
(C) The ellipsoid is not coincident with the geoid.
(D) The mass of the earth is irregularly distributed and, therefore, the geoi is also irregular.

Solution: The geoid is defined by the earth's gravity. The unequal distributio of the mass of the earth causes irregularity in the force and direction of gravity consequently, the geoid is also irregular.

Answer is (D)

9. What is the vertical datum most often used in the U.S.?

(A) the U.S. Standard Datum
(B) the IGL Datum
(C) the NAP Datum
(D) the National Geodetic Vertical Datum of 1929

Solution: The Sea Level Datum of 1929 was renamed in May 1976; now it i called the National Geodetic Vertical Datum of 1929. One reason for the ne name is that the datum does not accurately represent mean sea level or an equipotential surface. The datum was based on observed heights of mean se level at 26 tidal stations, 21 in the U.S. and 5 in Canada, along the Atlantic, Gul and Pacific coasts. Additionally, more than 100,000 km of leveling establishe benchmarks throughout the U.S.

Answer is (D)

10. In considering surveys of a very large extent, can a level surface at mean sea level be considered parallel with a level surface 100.0 m above mean sea level?

(A) No, a level surface is perpendicular everywhere to the direction of gravity.
(B) Yes, every level surface is parallel to every other level surface.
(C) No, level surfaces that are 100.0 m apart at the equator would be approximately 99.5 m apart at the poles, so they cannot be considered parallel.
(D) Both A and C are true.

Solution: While level surfaces may be considered parallel for surveying a limited area, in fact, they are not. The geoid, a close representation of a level surface, is not only discontinuous over variations of density in the earth, but is subject to a general convergence as well. The relative distance between two level surfaces will decrease from the equator to the poles as a consequence of the increase in gravity.

Answer is (D)

MAP PROJECTIONS

11. Which of the following best defines map projection?

(A) the systematic transformation of state plane coordinates into Cartesian coordinates
(B) the rigorous mathematical transference of points on a spheroid to the points on a plane surface
(C) the function of relating an ellipsoid of revolution to the geoid
(D) the mathematical development of the graticule representing the geographical lines on a globe

Solution: The earth is curved, and maps are flat. If the graticule, or grid, of a map is to correspond with the parallels of latitude and meridians of longitude on the terrestrial surface, an orderly method of transference must be adopted. There are many map projections that accomplish this; each has unique characteristics.

Answer is (B)

12. Which of the following map projections is used as the basis of state plane coordinate systems in the U.S.?

(A) the Lambert projection
(B) the oblique transverse Mercator projection
(C) the transverse Mercator projection
(D) A, B, and C

Solution: The Lambert projection and the transverse Mercator projections form the foundation for nearly all of the state plane coordinate systems. However, Zone 1 of the Alaska system is based on an oblique transverse Mercator projection.

Answer is (D)

13. Which of the following best describes a developable surface?

(A) a portion of the earth's surface that corresponds to mean sea level
(B) a polar gnomonic map projection
(C) a plane figure upon which the information from a spheroid may be projected and then flattened without gross distortion or tearing
(D) a surface that a map of a portion of the earth's surface may be drawn upon without any distortion

Solution: The difficulty of representing the earth's surface on a flat map can be solved by projecting the information from an ellipsoidal representation of the earth onto some surface that may be unrolled or developed, such as a cone or cylinder. These conic sections (if a cylinder may be so considered) can each be cut along an element and neatly flattened, minimizing distortion. Distortion can never be completely eliminated, however.

Answer is (C)

14. What characteristic of a map projection preserves the shape of small figures from the ellipsoid to the map?

(A) propagation
(B) standardization
(C) spatiality
(D) conformality

Solution: Some map projections allow for the angles between two curves on the spheroidal surface to be preserved on the map. This property, known as *conformality*, provides for the maintenance of elementary shapes on the map.

Answer is (D)

15. A rhumb line, not a cardinal line, has been drawn between two points on a map produced from a Mercator projection. Does the line represent the shortest distance between those two points on the earth?

(A) No, while the rhumb line would appear straight on a Mercator projection, it would not follow the arc of a great circle.
(B) Yes, the described loxodrome would follow the shortest path between two points on a conformal projection.
(C) No, the rhumb line would follow the arc of a great circle on the Mercator projection and therefore would have a constant geodetic direction.
(D) Yes, the rhumb line would follow the arc of a great circle on the Mercator projection.

Solution: A rhumb line, or loxodrome, appears to be a straight line on the Mercator projection. It crosses each meridian at a constant angle, but does not follow a great circle—the shortest path between two points.

Answer is (A)

16. Which of the following curves appears as a straight line on a gnomonic map projection?

(A) the parallels of latitude
(B) rhumb lines
(C) the arcs of great circles
(D) the arcs of small circles

Solution: The arc of a great circle will be a straight line on a gnomonic map projection, which is generated on a plane tangent to the ellipsoid at a single point.

Answer is (C)

THE LAMBERT PROJECTION

17. Which of the following terms may be correctly applied to the Lambert map projection that is used as the foundation for state plane coordinates in the U.S.?

(A) conformal and conic
(B) azimuthal and equal-area
(C) cylindrical and conformal
(D) conic and sinusoidal

Solution: Conformality is the characteristic of the Lambert projection that preserves the shape of small figures as they are transformed from the ellipsoid to the mapping plane. The Lambert projection is a conic projection; that is, the developable surface employed is imagined to be a cone.

Answer is (A)

18. The Lambert projection employs the Greek letter θ as the symbol for what important quantity?

(A) the scale factor
(B) the standard parallel
(C) the central meridian
(D) the mapping angle

Solution: The Greek letter θ is used in the Lambert projection to indicate the mapping angle.

Answer is (D)

19. The mapping plane used in the Lambert projection is often called a *secant cone*. Which of the following characteristics describes such a developable surface?

(A) The central axis of the cone is imagined to be a projection of the polar axis of the earth.
(B) The bottom of the cone is imagined to terminate at the earth's equator.
(C) The relationship between the altitude of the cone and its radius is always 1.55.
(D) The cone is imagined to intersect the ellipsoid along two standard parallels.

Solution: The secant cone intersects the ellipsoid at two lines of exact scale. This design allows the scale distortion to be minimized. Most of the mapping plane is actually between the standard parallels and, therefore, beneath the surface of the ellipsoid.

Answer is (D)

20. In a Lambert projection, the scale factor varies with which of the following quantities?

(A) the longitude
(B) the elevation
(C) the latitude
(D) the mapping angle

Solution: The scale factor in a Lambert projection is dependent on the latitude of a position.

Answer is (C)

21. The Lambert projections used as the basis of state plane coordinates in the U.S. employ two lines of strength, or standard parallels. Which of the following statements is true of the scale along a standard parallel in such a projection?

(A) The scale along a standard parallel is the smallest in the zone of a Lambert projection.

(B) The scale along a standard parallel is variable in the Lambert projection

(C) The scale along a standard parallel is the largest in the zone of a Lambert projection.

(D) The scale along a standard parallel in a Lambert projection is 1.

Solution: The standard parallels of a secant Lambert projection may be visualized as the intersection of the plane of the cone with the surface of the spheroid Any distance along a standard parallel would be exactly the same on the spheroid as on the map.

Answer is (D)

22. Meridians of longitude and parallels of latitude are represented differently in the Lambert projection. Which of the following statements best describes the difference between these two categories of lines?

(A) Meridians of longitude are represented by straight lines converging at the pole; parallels of latitude are represented by the arcs of concentric circles

(B) Meridians of longitude are represented by the arcs of concentric circles parallels of latitude are represented by straight lines.

(C) Parallels of latitude are represented by complex curves; meridians of longitude are represented by unequally spaced straight lines.

(D) Parallels of latitude are represented by curves concave toward the pole meridians of longitude are represented by complex curves concave toward the central meridian.

Solution: The conic Lambert projection represents parallels of latitude as arc of nearly equally spaced concentric circles. Meridians of longitude are shown a straight lines converging at the pole.

Answer is (A)

23. How many states use the Lambert conformal projection exclusively as the basis of their state plane coordinate systems?

(A) 18 states
(B) 20 states
(C) 28 states
(D) 33 states

Solution: Twenty-eight states use the Lambert projection exclusively as the basis of their state plane coordinate systems—Arkansas, California, Colorado, Connecticut, Iowa, Kansas, Kentucky, Louisiana, Maryland, Massachusetts, Minnesota, Montana, Nebraska, North Carolina, North Dakota, Ohio, Oklahoma, Oregon, Pennsylvania, South Carolina, South Dakota, Tennessee, Texas, Utah, Virginia, Washington, West Virginia, and Wisconsin.

Answer is (C)

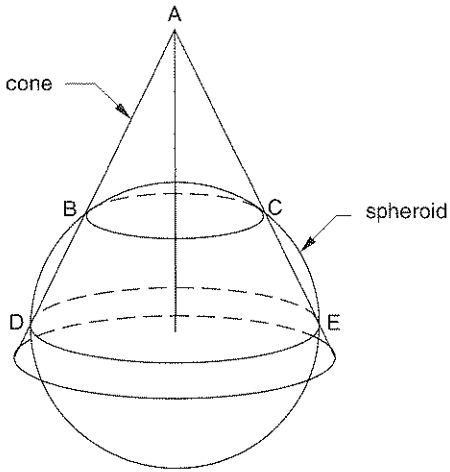
24. State plane coordinate systems based on Lambert conformal projections have been designed to minimize the distortion of grid lengths with respect to geodetic lengths. Where coverage by a single grid with a latitudinal extent of 158 statute miles is possible, which ratio correctly represents the maximum difference between the grid and geodetic lengths of a line?

(A) 1:79,000
(B) 1:40,000
(C) 1:5000
(D) 1:10,000

Solution: The target for maximum scale distortion is 1 part in 10,000. States that extend less than 158 mi north-south meet this goal. For example, the maximum scale in the Connecticut Coordinate System is about 1:40,000 along its southern and northern boundaries. On the other hand, the coverage of Texas by five zones was accomplished only by allowing distortion to go slightly more than 1:10,000.

Answer is (D)

The following two problems refer to the illustration shown.



25. Only part of the cone represented in the illustration is actually used in the developed Lambert projection. What is the typical portion included in a zone?

(A) Three-fifths of the zone is made up of the area of the cone within the spheroid—that is, between the standard parallels. One-fifth of the zone is the area north of the northerly standard parallel, and one-fifth is south of the southerly standard parallel.

(B) Only the portion of the cone within the spheroid is included in the zone.

(C) All of the area of the cone outside of the standard parallels BC and DE is included in the zone.

(D) Four-sixths of the zone is made up of the area of the cone within the spheroid—that is, between the standard parallels. One-sixth of the zone is the area north of the northerly standard parallel (BC) and one-sixth is south of the southerly standard parallel (DE).

Solution: Most frequently, the area between the standard parallels is approximately 105 mi, or four-sixths of the total zone. In other words, two-thirds of the area mapped is within the spheroid. One-sixth, or about 26 mi of the zone, is north of the northerly standard parallel and one-sixth is south of the southerly standard parallel. These latter areas are outside or above the spheroid.

Answer is (D)

26. What is the usual alignment of the axis of the cone in the illustration?

(A) The axis of the cone is perpendicular to the plane of the ecliptic.
(B) The axis of the cone is coincident with an extension of the earth's axis of rotation.
(C) The axis of the cone is arbitrary and has no typical arrangement.
(D) The axis of the cone is coincident with an extension of the earth's magnetic pole.

Solution: The usual alignment of the axis of the imaginary cone used in the Lambert projection is along the extension of the earth's polar axis.

Answer is (B)

THE TRANSVERSE MERCATOR PROJECTION

27. Two mapping projections have been used in the development of nearly all of the state plane coordinate systems in the U.S.; one is the Lambert projection and the other is the transverse Mercator projection. What is the significance of the word *transverse* in the name of the transverse Mercator projection?

(A) It refers to the characteristic representation of a line of constant azimuth on the spheroid as a straight line.
(B) It is used to indicate that the axis of the cylinder employed in the projection is perpendicular to the axis of the spheroid.
(C) It is used because the only straight line on the projection is the central meridian.
(D) The cylindrical surface used in the transverse Mercator projection is tangent to the spheroid at the equator.

Solution: The transverse Mercator projection is distinct from the Mercator projection in that the axis of the cylindrical surface is imagined to be perpendicular to the axis of the spheroid.

Answer is (B)

28. Which of the following statements describes the treatment of scale in the Lambert and the transverse Mercator projections?

(A) The Lambert projection scale is variable with respect to changes in longitude; in the transverse Mercator projection, it varies with respect to latitude.

(B) In the transverse Mercator projection, the lines of exact scale are meridians of longitude; in the Lambert projection, they are called *standard parallels of latitude*.

(C) The transverse Mercator projections used as the foundation of state plane coordinate systems are designed to limit scale distortion to 1 part in 20,000; in Lambert-based systems, distortion is limited to 1 part in 10,000.

(D) The scale is constant up and down the central meridian of a transverse Mercator projection; it is variable along the central meridian of a Lambert projection.

Solution: The variation of the scale in a transverse Mercator projection is based on changes in longitude, rather than latitude. The scale is constant along the central meridian of a transverse Mercator projection.

Answer is (D)

29. The grid imposed on a transverse Mercator projection indicating a particular state plane coordinate system is composed of straight lines. However, the geographical lines—that is, the parallels of latitude and meridians of longitude—are not straight. What symbol is used to indicate the angle between grid north through a particular point and the meridian of longitude through the same point?

(A) ϕ
(B) θ
(C) $\Delta\alpha$
(D) λ

Solution: The symbol used in the transverse Mercator projection to indicate the mapping angle, or convergence, is $\Delta\alpha$.

Answer is (C)

30. The cylinder is the developable surface employed in the transverse Mercator projection. Which of the following statements correctly describes the configuration of the cylinder used in the state plane coordinate systems based on the transverse Mercator projection?

(A) The axis of the cylinder is imagined to be perpendicular to the rotational axis of the spheroid; it intersects the spheroid along two meridians of longitude.

(B) The axis of the cylinder is imagined to be perpendicular to the rotational axis of the spheroid; it intersects the spheroid along two ellipses that are equally distant and parallel with the plane of the central meridian.

(C) The axis of the cylinder is imagined to be perpendicular to the rotational axis of the spheroid; it is tangent to the spheroid along the central meridian.

(D) The axis of the cylinder is imagined to be perpendicular to the rotational axis of the spheroid; it is tangent to the spheroid along the equator.

Solution: The cylinder used to generate a transverse Mercator-based state plane coordinate system intersects the spheroid along two lines that describe ellipses on the spheroid. The lines are the lines of strength of the developed map. The scale factor is 1 along these lines; they are equidistant from, and parallel with, the central meridian of the zone.

Answer is (B)

31. Which of the following statements correctly describes some of the features of the projection known as the universal transverse Mercator (UTM)?

(A) UTM employs 60 zones; each includes 6° of longitude.

(B) The foundation of UTM is the Jacobi reference ellipsoid; the coordinates in each zone are expressed in meters.

(C) The foundation of UTM includes five different reference ellipsoids; the system is used to map from 80°Nϕ to 80°Sϕ.

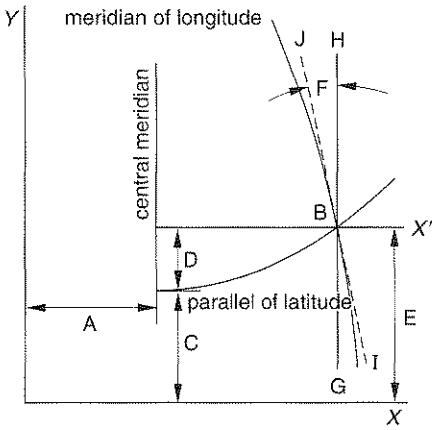
(D) Both A and C are true.

Solution: The UTM projection provides a worldwide system of coordinates in meters. Five different reference ellipsoids are used. They are the Clarke spheroids of 1866 and 1880, the international ellipsoid, the Bessel ellipsoid, and the Everest ellipsoid. The Clarke spheroid of 1866 is presently used in North America.

Answer is (D)

STATE PLANE COORDINATE SYSTEMS

The following six problems refer to the illustration shown.



The illustration shows the geometrical relationship between some of the significant lines comprising a state plane coordinate grid based on a transverse Mercator projection.

32. Considering NAD 27 state plane coordinate positions derived from a transverse Mercator projection, what is the arbitrary constant represented by the letter A?

(A) 500,000 ft
(B) 300,000 ft
(C) 700,000 ft
(D) 152,400.3 m

Solution: A constant of 500,000 ft is currently used in all of the states employing the transverse Mercator projection. This arbitrary constant between the Y-axis and the central meridian of each zone prevents negative X values in the coordinates of NAD 27. However, as states convert coordinate positions to NAD 83 this constant will be changed.

Answer is (A)

33. Considering NAD 83 state plane coordinate positions derived from a transverse Mercator projection, what arbitrary constant, represented by the letter A, will be used?

(A) 200,000 m, 400,000 m, 600,000 m, and 800,000 m
(B) 213,360 m, 850,000 m, and 165,000 m
(C) 100,000 m, 300,000 m, and 900,000 m
(D) all of the above and more

Solution: The conversion of state plane coordinate positions to NAD 83 will involve a change in the arbitrary distance between the Y-axis and the central meridian of transverse Mercator zones. Recommendations for the change include the expression of the now standard 500,000 ft in meters, a change from 500,000 ft to 300,000 ft, and a change from 500,000 ft to 700,000 ft.

The objective is to avoid any confusion of NAD 27 with NAD 83 coordinate positions. The conversion of feet to meters is a bit problematic since some states use the U.S. survey foot, which is 0.3048006 of a meter, while some prefer to use the international foot, which is 0.3048000 of a meter. No single value is used universally, as was the 500,000-ft value. All of the quantities shown have been proposed in various states and others are likely to be proposed.

Answer is (D)

34. Line GH is parallel with the central meridian, the dashed line IJ is not parallel, and the angle between them is symbolized by $\Delta\alpha$. Which of the following statements would be correct for an observer standing at station B and facing geodetic north?

(A) The observer would find line IJ on the left of geodetic north and line GH on the right.
(B) The observer would find line GH on the left of geodetic north and line IJ on the right.
(C) The observer would find both line GH and line IJ on the left of geodetic north.
(D) The observer would be looking along line IJ and line GH would be on the right.

Solution: The line labeled IJ represents geodetic north at station B. Line GH represents grid north and is parallel to the central meridian.

Answer is (D)

35. Which of the following labels are commonly used for the distances labeled C, D, and E?

(A) C = Y_o
D = $\Delta\phi$
E = Y-coordinate

(B) C = $\pm ab$
D = $\Delta\phi$
E = $\Delta\lambda$

(C) C = H
D = V
E = Y_o

(D) C = Y-coordinate
D = Y_o
E = $\Delta\phi$

Solution: The symbols most consistently used in the state plane coordinate systems based on the transverse Mercator projection are C = Y_o, D = $\Delta\phi$, and E = Y-coordinate. The Y-coordinate of station B is the sum of Y_o, called tabular Y, and the quantity $\Delta\phi$. Note that the parallel of latitude through station B intersects the central meridian somewhat to the south of the line labeled X'.

Answer is (A)

36. Which of the distances is known as a *spheroidal perpendicular* and is equal to the expression $H\Delta\lambda'' \pm ab$?

(A) A
(B) X'
(C) C
(D) D

Solution: The spheroidal perpendicular is the distance from the central meridian along a grid line to the station under consideration, and is usually symbolized by X'. The length of the spheroidal perpendicular is

$$K\left(S_g + \frac{S_g^3}{6\rho^2}\right)$$

S_g is the actual geodetic length of the perpendicular, ρ is the average radius of the spheroid, and K is the scale factor.

Answer is (B)

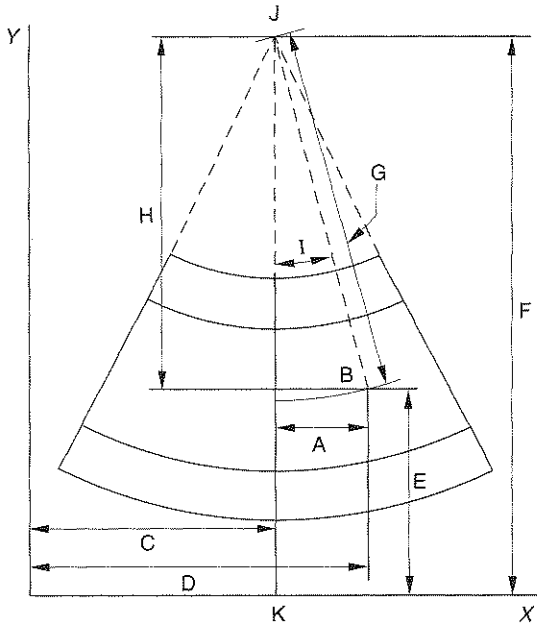
37. Which of the following expressions is correct for the angular value symbolized by the letter F?

(A) $\pm\Delta\alpha$
(B) $\Delta\alpha'$
(C) $+\Delta\alpha$
(D) $-\Delta\alpha$

Solution: East of the central meridian of a zone, $\Delta\alpha$, the convergence (or mapping angle) is positive (+); west of the central meridian, the value is negative (−).

Answer is (C)

The following six problems refer to the illustration shown.



The illustration shows the geometrical relationship between some of the significant lines comprising a state plane coordinate grid based upon a Lambert projection.

38. Considering NAD 27 state plane coordinate positions derived from a Lambert projection, what is the arbitrary constant represented by the letter C?

(A) 1,000,000 m
(B) 500,000 ft
(C) 100,000 ft
(D) 2,000,000 ft

Solution: A constant of 2,000,000 ft is currently used in all of the states employing the Lambert projection. This arbitrary constant between the Y-axis and the central meridian of each zone prevents negative X values in the coordinates of NAD 27. However, as states convert coordinate positions to NAD 83, this constant will be changed.

Answer is (D)

39. Which of the following symbols most completely describes the angular value labeled I?

(A) $-\theta$
(B) $+\theta$
(C) θ
(D) θ'

Solution: The usual symbol for the mapping angle, or convergence, in the Lambert projection is the Greek letter θ. It is considered to be positive to the east of the central meridian and negative to the west.

Answer is (B)

40. Which of the following expressions is commonly used to represent the distance labeled A?

(A) X'
(B) $R\cos\theta$
(C) R_b
(D) both A and B

Solution: The distance labeled A is known as X'. It can also be represented by the expression $R\sin\theta$.

Answer is (A)

41. Using the most typical symbols, which of the formulas given is equivalent to F − E = H?

(A) $R - X = Y$
(B) $R\cos\theta - Y = R_b$
(C) $P - R\sin\theta = C$
(D) $R_b - Y = R\cos\theta$

Solution: The distance labeled E is the Y-coordinate of station B. The distance from the apex of the cone to the grid line through station B is equal to $R\cos\theta$. The distance labeled F is usually symbolized R_b.

Answer is (D)

42. What is the name usually given to the line labeled JK in the illustration?

(A) the Y-axis
(B) ϕ_o
(C) the central meridian
(D) λ_1

Solution: The central meridian is the name usually given to the longitudinal line chosen near the center of the zone.

Answer is (C)

43. The distance labeled D represents the X-coordinate of station B, and the distance labeled E is its Y-coordinate. Which of the following formulas correctly represents these coordinates?

(A) $X = 2{,}000{,}000 - X'$
$Y = R_b - R\cos\theta$
(B) $X = R\sin\phi + C$
$Y = R_b - R\cos\theta$
(C) $X = 2{,}000{,}000 + X'$
$Y = R\cos\theta - R_b$
(D) $X = R\sin\theta + C$
$Y = R_b - R\cos\theta$

Solution: The value of the mapping angle, θ, is sometimes positive and sometimes negative. Therefore, the correct formulas for the X- and Y-coordinates in the Lambert projection are

$$X = R\sin\theta + C$$
$$Y = R_b - R\cos\theta$$

Answer is (D)

44. Some elements are constant for each zone of state plane coordinates based on a Lambert projection, and some vary with each station. Which elements are constant?

(A) Each zone has a unique origin, a constant R_b, and a distance of 2,000,000 ft from the Y-axis to the central meridian.

(B) Each zone has a unique central meridian, a constant X', and a distance of 2,000,000 ft from the Y-axis to the central meridian.

(C) Each zone has a unique central meridian, a constant R, and a distance of 2,000,000 ft from the X-axis to the central meridian.

(D) Each zone has a unique central meridian, a constant R_b, and a distance of 2,000,000 ft from the X-axis to the central meridian.

Solution: The constant C is 2,000,000 ft for each zone, and each zone has a unique origin for its coordinates. A third constant value, known as R_b, is the distance from the apex of the cone to its X-axis.

Answer is (A)

45. The following lists are made up of variables from state plane coordinate systems based on the Lambert projection and the transverse Mercator projection. Which lists show only those variables that vary with latitude?

(A) Lambert: $\Delta\alpha$, H, V, and a
transverse Mercator: θ, R, scale factor, and b

(B) Lambert: ℓ, X', and θ
transverse Mercator: b, c, and Y_o

(C) Lambert: R, scale factor, and θ
transverse Mercator: Y_o, H, V, and a

(D) Lambert: Y_o, H, V, and a
transverse Mercator: R, scale factor, and θ

Solution: The variables that are dependent on latitude in the Lambert projection are: R, the radius from the apex of the cone to a given station; θ, the

convergence; and the scale factor. In a transverse Mercator projection, the geodetic multipliers H and V, Y_o, and the small correction a vary with the changes in latitude.

Answer is (C)

46. Most state plane coordinate systems are designed to limit scale distortion to 1 part in 10,000. If this goal has been achieved, within what limits should the scale factor fall?

(A) not less than 0.9999823;
not more than 1.0001787

(B) not less than 0.9999569;
not more than 1.0000839

(C) not less than 0.9994563;
not more than 1.0051020

(D) not less than 0.9999000;
not more than 1.0001000

Solution: Distortion of one part in 10,000 implies a scale factor of 0.9999 at the lower limit and 1.0001 at the upper limit. While some state plane coordinate zones may exceed these limits, the objective is to stay within them whenever possible.

Answer is (D)

47. Between the standard parallels of a state plane coordinate zone based on the Lambert projection, the scale factor remains within what specific limits?

(A) The scale factor is always greater than 1 between the standard parallels.

(B) The scale factor is always greater than 1 and less than 2 between the standard parallels.

(C) The scale factor is always greater than zero and less than 1 between the standard parallels.

(D) The scale factor is always less than zero between the standard parallels.

Solution: The developable surface of the mapping plane is below the spheroid between the standard parallels of a Lambert secant projection. Therefore, the distances on the spheroid are always reduced when they are projected onto the mapping plane. In other words, the scale factor is always less than 1 between the standard parallels.

Answer is (C)

48. In state plane coordinate systems derived from a transverse Mercator projection, what value is the scale factor based on?

(A) $\Delta\lambda''$
(B) $\Delta\phi$
(C) Y-coordinate
(D) X'

Solution: The distance from a given point to the central meridian, also known as the *spheroidal perpendicular*, is the basis for the derivation of the scale factor in the transverse Mercator projection. The spheroidal perpendicular is symbolized by X'.

Answer is (D)

49. Considering a state plane zone based on the transverse Mercator projection, what are the limits of the scale factor outside the lines of exact scale?

(A) The scale factor is always greater than 1 outside the lines of exact scale.
(B) The scale factor is always 1 outside the lines of exact scale.
(C) The scale factor is always less than zero outside the lines of exact scale.
(D) The scale factor is always greater than zero and less than 1 outside the lines of exact scale.

Solution: The cylinder is above the spheroid outside the lines of its intersection, which are the lines of exact scale. The distances on the spheroid are increased

when they are projected onto the developable surface. Therefore, the scale factor is greater than 1.

Answer is (A)

50. Which factor is used to directly reduce distances measured on the topographic surface of the earth to the state plane?

(A) the scale factor
(B) the K factor
(C) the sea-level factor
(D) the combination factor

Solution: The combination, or grid, factor is used to move directly from the earth's surface to the state plane in one step. This factor is the product of the scale, or K, factor and the sea-level factor.

Answer is (D)

51. What approximation of the radius of the reference ellipsoid has traditionally been used as the basis for calculating the sea-level factor?

(A) 24,000,000 ft
(B) 20,906,000 ft
(C) 6,378,206.4 m
(D) 20,925,620 ft

Solution: The mean radius of the spheroid over the coterminous U.S. has been considered as 20,906,000 ft for many years. This approximation is adequate for many applications, although the actual figure varies as much as 50,000 ft across the 48 states.

Answer is (B)

52. It is sometimes appropriate to apply a weighted mean scale factor to a line of 5 mi or more. Which of the following circumstances would warrant such a refinement?

(A) a long, predominantly east-west line on the Lambert projection or a long, predominantly north-south line on the transverse Mercator projection

(B) a long, predominantly east-west line on the transverse Mercator projection or a long, predominantly north-south line on the Lambert projection

(C) a long line near the center of the zone on the transverse Mercator projection and a similar line on the Lambert projection

(D) a long line near the edge of the zone on the transverse Mercator projection and a similar line on the Lambert projection

Solution: The scale factor varies with the latitude in the Lambert projection and with the longitude in the transverse Mercator projection. Large changes in these elements of the scale factor, especially in long lines, may require the use of the weighted mean in calculating the scale factor.

Answer is (B)

53. There is a parallel of latitude known as ϕ_o in each zone based on the Lambert projection. Which of the following statements correctly identifies some of the special characteristics of this parallel of latitude?

(A) All of the geodetic lines in the zone are slightly concave toward this parallel of latitude.

(B) The scale factor ratio is lowest along this parallel of latitude.

(C) The latitude of this line is equidistant from each of the standard parallels.

(D) Both A and B are true.

Solution: The ϕ_o latitude is near, but not precisely at, the center of the zone. The latitude of the line can be found as $\sin^{-1} \ell$. The scale factor ratio reaches its minimum along the ϕ_o line, and the geodetic lines projected onto the state plane in the zone are concave toward it.

Answer is (D)

54. Which of the following symbols is normally used to indicate the second-term correction applied to the direction of long lines in the Lambert state plane coordinate system?

(A) ω
(B) θ'
(C) $\Delta\alpha'$
(D) σ

Solution: In the Lambert projection, the most common symbol for the second-term correction is θ'. The second term is a compensation applied to the direction of long lines on the state plane that would otherwise retain a slight curvature toward the center of the zone—that is, toward the parallel of latitude known as ϕ_o.

Answer is (B)

55. What is the formula used to convert a geodetic azimuth to a grid azimuth in the Lambert conformal projection when a relatively short line is involved?

(A) grid azimuth = geodetic azimuth + θ
(B) grid azimuth = geodetic azimuth − θ
(C) grid azimuth = θ − geodetic azimuth
(D) grid azimuth = geodetic azimuth + $\theta - \theta'$

Solution: The formula for conversion of a geodetic azimuth to a grid azimuth is

$$\text{grid azimuth} = \text{geodetic azimuth} - \theta$$

This conversion ignores the effect of the second term since a relatively short line is involved.

Answer is (B)

56. Which of the following expressions is equal to the mapping angle in the transverse Mercator projection?

(A) $X' + 500{,}000$

(B) $Y_o + V\left(\frac{\Delta\lambda''}{100}\right)^2 \pm c$

(C) $H\Delta\lambda'' \pm ab$

(D) $\Delta\lambda'' \sin\phi + g$

Solution: The expression $\Delta\lambda'' \sin\phi + g$ is equivalent to $\Delta\alpha$, the mapping angle, in the transverse Mercator projection. The change in longitude in seconds, $\Delta\lambda''$, multiplied by the sine of the latitude is adjusted by the small quantity g, normally found in published tables.

Answer is (D)

57. What is the relationship between the mapping angle, θ, and the distance from the central meridian for a station in a zone of a state plane coordinate system based upon the Lambert projection?

(A) The absolute value of θ is always larger than the difference in longitude from the central meridian.

(B) θ is not always the same for a specific meridian of longitude.

(C) The absolute value of θ is always the same as the difference in longitude from the central meridian.

(D) The absolute value of θ is always smaller than the difference in longitude from the central meridian.

Solution: θ is positive to the east of the central meridian and negative to the west. Its absolute value is always less than the change in longitude from the central meridian.

Answer is (D)

58. Which of the following expressions is equal to the mapping angle, θ, in a state plane coordinate system based on the Lambert projection?

(A) $\Delta\lambda\ell$

(B) $\tan^{-1}\left(\dfrac{X'}{R_b - y}\right)$

(C) $\sin^{-1}\left(\dfrac{R_b - y}{R}\right)$

(D) both A and B

Solution: The following formulas are usually given in state plane projection tables.

$$\Delta\lambda = \frac{\theta}{\ell}$$

$$\theta = \Delta\lambda\ell$$

$$\tan\theta = \frac{X'}{R_b - y}$$

$$\theta = \tan^{-1}\frac{X'}{R_b - y}$$

Answer is (D)

SHORT GEODETIC CALCULATIONS

59. The geodetic azimuth (from south) from station Telo to station Klamath is $186°15'22''.21$, and the mapping angle at station Telo is $+00°40'21''.85$. What is the grid azimuth from station Telo to station Klamath if the second term is not considered?

(A) $186°55'44''.06$
(B) $06°55'44''.06$
(C) $05°35'00''.36$
(D) $185°35'00''.36$

Solution: Since the geodetic azimuth is measured clockwise from south, subtract 180° to find the geodetic azimuth from north.

$$\begin{array}{r} 186^\circ 15' 22.''21 \\ \underline{-180^\circ 00' 00.''00} \\ 06^\circ 15' 22.''21 \end{array}$$

The formula for finding the grid azimuth of short lines is

$$\begin{aligned} \text{grid azimuth} &= \text{geodetic azimuth} - \theta \\ &= 06^\circ 15' 22.''21 - (+00^\circ 40' 21.''85) \\ &= 05^\circ 35' 00.''36 \end{aligned}$$

Answer is (C)

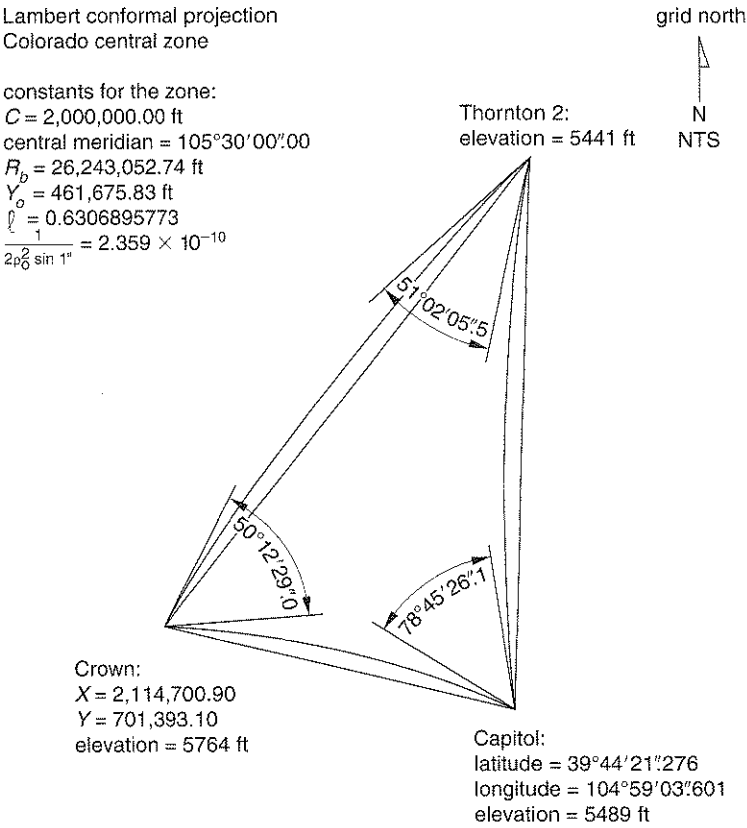
60. The grid azimuth from station Exeter to station Abott is 212°32′14.″1, and the mapping angle at station Exeter is +00°27′45.″2. What is the geodetic azimuth from station Abott to station Exeter if the second term is not considered?

(A) 212°59′59.″3
(B) 32°59′59.″3
(C) 212°04′28.″9
(D) insufficient information given

Solution: There is not enough information given to answer this question. While the grid azimuth will be precisely 180° different at station Exeter (212°32′14.″1) from its value at station Abott (32°32′14.″1), the same cannot be said of the geodetic azimuth. Without the value of the mapping angle at station Abott or some information from which it may be derived, the question cannot be answered.

Answer is (D)

The following seven problems refer to the illustration shown.



Note that the concave lines connecting the three stations indicate the curvature of long lines, such as those in the illustration, on the state plane. The angles shown are observed values.

61. The following values are taken from the *Plane Coordinate Projection Tables—Colorado*, published by the U.S. Government Printing Office.

Latitude	R (ft)	Tabular Difference for 1″ of Latitude (ft)
39°41′	25,569,283.24	101.18217
42′	25,563,212.31	101.18283
43′	25,557,141.34	101.18350
44′	25,551,070.33	101.18400
45′	25,544,999.29	101.18467

What is the appropriate R value for the station labeled Capitol?

(A) 25,547,152.08 ft
(B) 25,541,081.04 ft
(C) 25,554,988.58 ft
(D) 25,548,917.54 ft

Solution: The R value for a particular station is the distance from the apex of the cone to that station in feet. One way to derive its value is to use the tabular difference for 1″ of latitude given.

First, find the difference in seconds of latitude from the next smaller minute given.

$$\begin{aligned} \text{latitude of station Capitol} &= 39°44'21.''276 \\ &\underline{-39°44'00.''000} \\ \text{difference in seconds} &= 21.''276 \end{aligned}$$

Next, multiply this difference by the tabular difference for 1″ of latitude to find the difference between the R value at latitude 39°44′00.″000 and at latitude 39°44′21.″276.

$$(21.''276)(101.18400 \text{ ft}) = 2152.79 \text{ ft}$$

Finally, subtract this difference from the R value given for the next smaller latitude.

$39°44'$	25,551,070.33 ft	101.18400 ft
	$\underline{-2152.79 \text{ ft}}$	
$39°44'21.''276$	25,548,917.54 ft	
$39°45'$	25,544,999.29 ft	101.18467 ft

Answer is (D)

62. What is the value of θ for the station labeled Capitol?

(A) $-00°20'10.''92$
(B) $+00°19'35.''35$
(C) $-00°30'56.''39$
(D) $+00°19'30.''81$

Solution: There is more than one way to calculate the θ angle from the longitude of a station. One way is to use the following relationship.

$$\theta = \Delta\lambda\ell$$

The symbol $\Delta\lambda$ refers to the change in longitude from the central meridian of the zone, which in this case is from $105°30'00''$.

$$\begin{aligned} \text{central meridian} &= 105°30'00.''000 \\ \lambda \text{ of Capitol} &= \underline{-104°59'03.''601} \\ \Delta\lambda &= +00°30'56.''399 \end{aligned}$$

The symbol ℓ indicates the sine of the apex angle for the zone. It is shown in the illustration as 0.6306895773.

$$\begin{aligned} \theta &= \Delta\lambda\ell \\ &= (+00°30'56.''399)(0.6306895773) \\ &= +00°19'30.''81 \end{aligned}$$

Answer is (D)

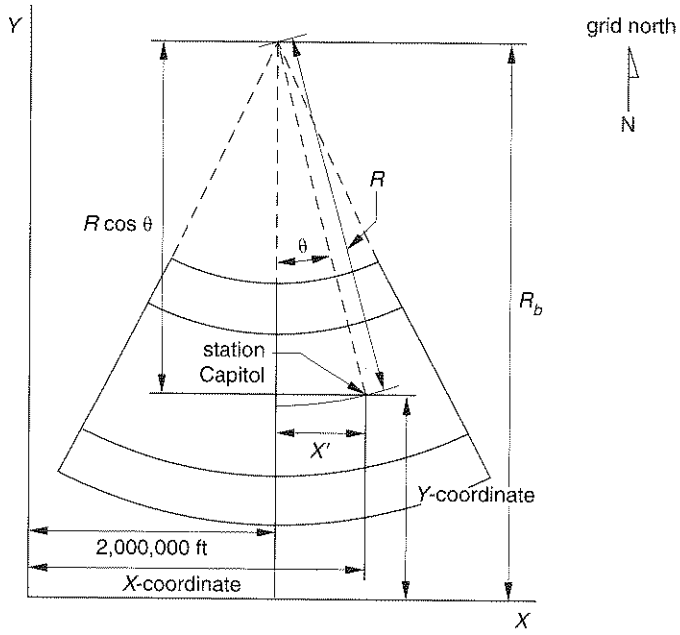
63. What is the X-coordinate of station Capitol?

(A) $X = 25{,}548{,}507.95$
(B) $X = 145{,}021.38$
(C) $X = 27{,}548{,}505.95$
(D) $X = 2{,}145{,}021.38$

Solution: The value of θ for station Capitol is $+00°19'30.''8115$, and the value of R is 25,548,917.54 ft. These two quantities are sufficient to find the value of the X-coordinate, using the following formula.

$$X = R\sin\theta + C$$

C is the constant 2,000,000 ft between the central meridian and the Y-axis. The following figure illustrates these relationships.



$$\begin{aligned} X &= R\sin\theta + C \\ &= (25{,}548{,}917.54 \text{ ft})(\sin +00°19'30.''8115) + 2{,}000{,}000 \text{ ft} \\ &= 145{,}021.38 \text{ ft} + 2{,}000{,}000 \text{ ft} \\ &= 2{,}145{,}021.38 \text{ ft} \end{aligned}$$

Answer is (D)

64. What is the Y-coordinate of station Capitol?

(A) $Y = 609{,}803.31$
(B) $Y = 694{,}546.79$
(C) $Y = 687{,}312.94$
(D) $Y = 706{,}091.46$

Solution: The value of θ for station Capitol is $+00°19'30.''8115$, and the value of R is 25,548,917.54 ft. These two quantities are sufficient to find the value of the Y-coordinate, using the following formula.

$$Y = R_b - R\cos\theta$$

R_b is the constant for the zone. The value of R_b for the central zone of Colorado is 26,243,052.74 ft, as shown in the illustration of stations Crown, Capitol, and Thornton 2.

$$\begin{aligned} Y &= R_b - R\cos\theta \\ &= 26{,}243{,}052.74 \text{ ft} - (25{,}548{,}917.54 \text{ ft})(\cos +00°19'30.''8115) \\ &= 26{,}243{,}052.74 \text{ ft} - 25{,}548{,}505.95 \text{ ft} \\ &= 694{,}546.79 \text{ ft} \end{aligned}$$

Answer is (B)

65. What is the angular spherical excess in the figure created by Crown, Capitol, and Thornton 2?

(A) 3.1 sec
(B) 0.4 sec
(C) 1.2 sec
(D) 0.8 sec

Solution: The area of the figure can be used to find the spherical excess. Each 75.6 sq mi included in the figure corresponds to approximately 1 sec of spherical excess. The area of the figure can be found as follows.

Crown:

$$\begin{aligned} X &= 2{,}114{,}700.90 \\ Y &= 701{,}393.10 \end{aligned}$$

Capitol:

$$\begin{aligned} X &= 2{,}145{,}021.38 \\ Y &= 694{,}546.79 \end{aligned}$$

$$\text{distance from Crown to Capitol} = \text{distance } a = 31{,}083.81 \text{ ft [grid]}$$

$$\text{angle at Crown} = \text{angle B} = 50°12'29.''0$$

$$\text{angle at Capitol} = \text{angle C} = 78°45'26.''1$$

$$\text{angle at Thornton 2} = \text{angle A} = 51°02'05.''5$$

Therefore, the area of the figure is

$$\begin{aligned} \text{A} &= \frac{a^2 \sin \text{B} \sin \text{C}}{\sin \text{A}} \\ &= \frac{(31{,}083.81 \text{ ft})^2 (\sin 50°12'29.''0)(\sin 78°45'26.''1)}{\sin 51°02'05.''5} \\ &= \frac{(966{,}203{,}244.1 \text{ sq ft})(0.753628393)}{0.777528722} \\ &= \frac{728{,}158{,}198.5 \text{ sq ft}}{0.777528722} \\ &= 936{,}503{,}280.1 \text{ sq ft} \end{aligned}$$

Convert to square miles.

$$(936{,}503{,}280.1 \text{ sq ft})\left(\frac{1 \text{ sq mi}}{5280 \text{ sq ft}}\right) = 33.6 \text{ sq mi}$$

$$\text{spherical excess} = \frac{33.6 \text{ sq mi}}{75.6\,\dfrac{\text{sq mi}}{\text{sec}}}$$

$$= 0.4 \text{ sec}$$

Answer is (B)

66. What is the second-term, or θ', correction that would be applied at station Crown for the line Crown-Capitol?

(A) $-01.''2$
(B) $+00.''9$
(C) $+01.''7$
(D) $-00.''4$

Solution: The second-term correction is an angular value that is applied to long lines, generally lines of more than 5 mi in length. The line Crown-Capitol is nearly 6 mi long. The second-term correction is an accommodation of the residual curvature present in such lines even after the development of the mapping surface. The formula for finding the second-term correction in a Lambert projection is

$$\theta' = \left(\frac{X_2 - X_1}{2\rho_o^2 \sin 1''}\right)\left(Y_1 - Y_o + \frac{Y_2 - Y_1}{3}\right)$$

X_1, Y_1 and X_2, Y_2 are the coordinates of the beginning and the end of the line in question, respectively. Y_o is the Y-coordinate of the parallel of latitude near the center of the zone, which is a constant for each zone. The Y_o value for the central zone of Colorado is 461,675.83 ft.

The quantity $1/(2\rho_o^2 \sin 1'')$ is also a constant for each zone and is 2.359×10^{-10} in the central zone of Colorado. Substituting the appropriate values in the formula,

Crown:

$$\begin{aligned} X_1 &= 2{,}114{,}700.90 \\ Y_1 &= \phantom{2{,}}701{,}393.10 \end{aligned}$$

Capitol:

$$\begin{aligned} X_2 &= 2{,}145{,}021.38 \\ Y_2 &= \phantom{2{,}}694{,}546.79 \end{aligned}$$

$$\begin{aligned} \theta' &= \left(\frac{X_2 - X_1}{2\rho_o^2 \sin 1''}\right)\left(Y_1 - Y_o + \frac{Y_2 - Y_1}{3}\right) \\ &= \left(\frac{2{,}145{,}021.4 - 2{,}114{,}700.9}{2\rho_o^2 \sin 1''}\right) \\ &\quad \times \left(701{,}393.1 - 461{,}675.8 + \frac{694{,}546.8 - 701{,}393.10}{3}\right) \\ &= (30{,}320.5)(0.0000000002359)\left(239{,}717.3 + \frac{-6846.3}{3}\right) \\ &= (0.000007153)[239{,}717.3 + (-2282.1)] \\ &= (0.000007153)(237{,}435.2) \\ &= 01''.7 \end{aligned}$$

Answer is (C)

67. What is the second-term, or θ', correction that would be applied at station Crown for the line Capitol-Crown?

(A) $-0''.08$
(B) $-01''.8$
(C) $+01''.7$
(D) $+01''.9$

Solution: The calculation for finding the second-term correction for a long line can be simplified somewhat through expression of the numbers involved with exponents of 10. For example, the value of $(1/2\rho_o^2 \sin 1'')$ for the central zone

of Colorado can be expressed as 2.359×10^{-10} instead of 0.0000000002359. The coordinates of the points Crown and Capitol can be simplified in the same manner.

Capitol:

$$X_1 = 21.450 \times 10^5$$
$$Y_1 = 6.945 \times 10^5$$

Crown:

$$X_2 = 21.115 \times 10^5$$
$$Y_2 = 7.014 \times 10^5$$

Similarly, the constant Y_o can be expressed as 4.617×10^5.

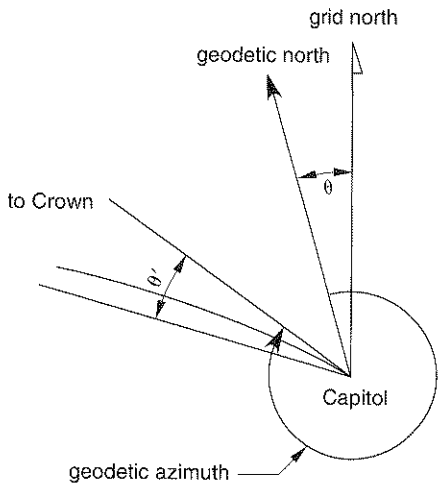
$$\begin{aligned}
\theta' &= \left(\frac{X_2 - X_1}{2\rho_o^2 \sin 1''}\right)\left(Y_1 - Y_o + \frac{Y_2 - Y_1}{3}\right) \\
&= \left(\frac{21.115 \times 10^5 - 21.450 \times 10^5}{2\rho_o^2 \sin 1''}\right) \\
&\quad \times \left(6.945 \times 10^5 - 4.617 \times 10^5 + \frac{7.014 \times 10^5 - 6.945 \times 10^5}{3}\right) \\
&= (-0.335 \times 10^5)(2.359 \times 10^{-10})\left(2.328 \times 10^5 + \frac{0.069 \times 10^5}{3}\right) \\
&= (-0.335 \times 10^5)(2.359 \times 10^{-10})(2.351 \times 10^5)
\end{aligned}$$

When multiplying, add exponents.

$$\begin{aligned}
\theta' &= (-0.335 \times 10^5)(5.546 \times 10^{-5}) \\
&= -01''.8
\end{aligned}$$

Answer is (B)

68. What is the geodetic azimuth from station Capitol to station Crown?



(A) 257°36′06.″5
(B) 257°16′35.″7
(C) 282°43′26.″1
(D) 283°02′58.″7

Solution: The grid azimuth from Capitol to Crown, 282°43′26.″1, may be found by inversing the grid coordinates.

Crown:

$$X = 2{,}114{,}700.90$$
$$Y = 701{,}393.10$$

Capitol:

$$X = 2{,}145{,}021.38$$
$$Y = 694{,}546.79$$

The mapping angle, θ, can be found from the expression

$$\theta = \Delta\lambda\ell$$

$\Delta\lambda$ is the change in longitude from the central meridian, and ℓ is a constant for the zone. The central meridian for Colorado is 105°30′00″, and the constant ℓ is 0.6306895773.

$$\begin{aligned}\theta &= \Delta\lambda\ell \\ &= (105°30'00''.0 - 104°59'03''.6)(0.6306895773) \\ &= (00°30'56''.4)(0.6306895773) \\ &= 00°19'30''.8\end{aligned}$$

The second-term, or θ', value is $-01''.8$, as derived in Problem 67.

$$\begin{aligned}\theta' &= \left(\frac{X_2 - X_1}{2\rho_o^2 \sin 1''}\right)\left(Y_1 - Y_o + \frac{Y_2 - Y_1}{3}\right) \\ &= \left(\frac{21.115 \times 10^5 - 21.450 \times 10^5}{2\rho_o^2 \sin 1''}\right) \\ &\quad \times \left(6.945 \times 10^5 - 4.617 \times 10^5 + \frac{7.014 \times 10^5 - 6.945 \times 10^5}{3}\right) \\ &= (-0.335 \times 10^5)(2.359 \times 10^{-10})\left(2.328 \times 10^5 + \frac{0.069 \times 10^5}{3}\right) \\ &= (-0.335 \times 10^5)(2.359 \times 10^{-10})(2.351 \times 10^5) \\ &= (-0.335 \times 10^5)(5.546 \times 10^{-5}) \\ &= -01''.8\end{aligned}$$

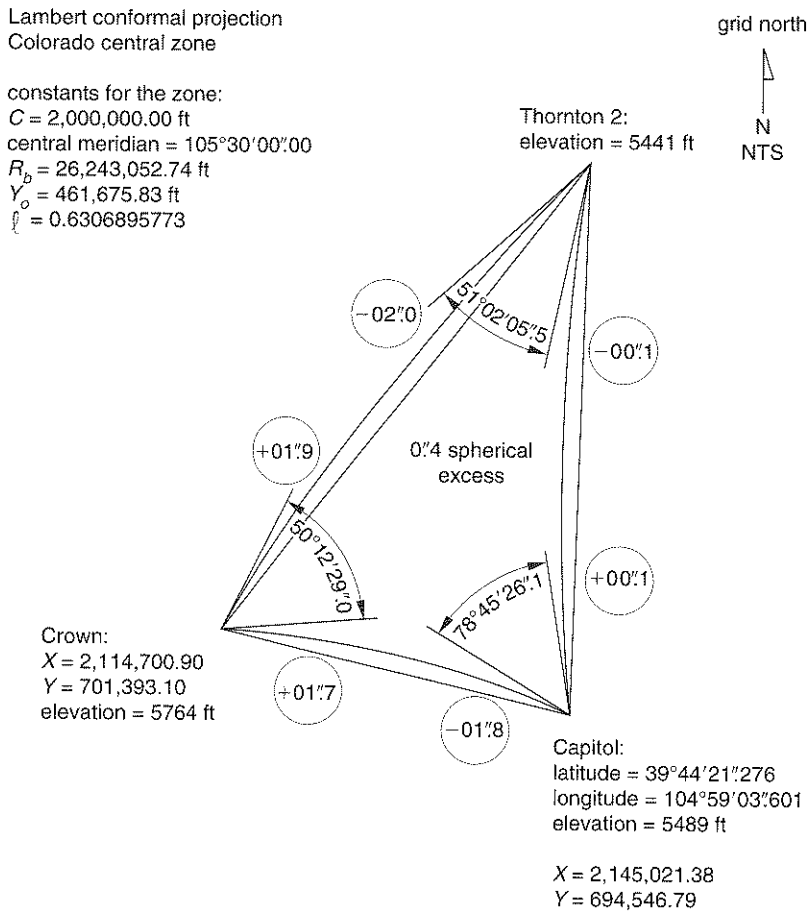
The formula for the geodetic azimuth of a long line when the grid azimuth is available is

$$\begin{aligned}\text{geodetic azimuth} &= \text{grid azimuth} + \theta - \theta' \\ &= 282°43'26''.1 + (+00°19'30''.8) - (-00°00'01''.8) \\ &= 283°02'58''.7\end{aligned}$$

Answer is (D)

69. In the illustration shown, the concave lines connecting the three stations indicate the curvature of long lines on the state plane. The angles shown are observed values. The angular values circled are the second-term, or θ', values.

Given this information, what are the correct values for the three interior angles of the figure after both second terms and spherical excess have been taken into consideration?



(A) $51°02'03.''6$
$78°45'28.''0$
$50°12'28.''8$

(B) $51°02'05.''5$
$78°45'26.''1$
$50°12'29.''0$

(C) $51°02'02.''3$
$78°45'26.''7$
$50°12'27.''5$

(D) $51°02'03.''5$
$78°45'27.''9$
$50°12'28.''7$

Solution: The second terms and the spherical excess are given in the illustration. Each angle may be corrected individually.

angle at Thornton 2:

$$\begin{aligned} \text{observed} &= 51°02'05.''5 \\ \theta' &= \quad +(-02.''0) \\ &\quad\ -(-00.''1) \\ \text{one-third of spherical excess} &= \quad -(+00.''13) \\ &\overline{\quad 51°02'03.''47} \end{aligned}$$

Rounded to the nearest tenth of a second, this value is $51°02'03.''5$.

angle at Capitol:

$$\begin{aligned} \text{observed} &= 78°45'26.''1 \\ \theta' &= \quad +(+00.''1) \\ &\quad\ -(-01.''8) \\ \text{one-third of spherical excess} &= \quad -(+00.''13) \\ &\overline{\quad 78°45'27.''87} \end{aligned}$$

Rounded to the nearest tenth of a second, this value is $78°45'27.''9$.

angle at Crown:

$$\begin{aligned} \text{observed} &= 50°12'29.''0 \\ \theta' &= \quad +(+01.''7) \\ &\quad\ -(+01.''9) \\ \text{one-third of spherical excess} &= \quad -(+00.''13) \\ &\overline{\quad 50°12'28.''67} \end{aligned}$$

Rounded to the nearest tenth of a second, this value is $50°12'28.''7$.

Answer is (D)

70. Given the value of ℓ for the Colorado central zone of 0.6306895773, what is the parallel of latitude for that zone along which the scale factor ratio is smallest?

(A) $39°06'03.''66$
(B) $38°27'00.''00$
(C) $39°45'00.''00$
(D) $39°33'21.''53$

Solution: The latitude crossing the region where the spheroid is farthest outside the cone is the line where the scale factor ratio is smallest. The ℓ value, a constant for each zone, is the sine of that latitude.

$$\sin^{-1}(0.6306895773) = 39°06'03.''66$$

Answer is (A)

71. The state plane coordinates of Thornton 2 in the Colorado central zone are

$$\begin{aligned} X &= 2{,}146{,}262.93 \\ Y &= 742{,}549.22 \end{aligned}$$

The constants for the zone are

$$\begin{aligned} C &= 2{,}000{,}000 \text{ ft} \\ \text{central meridian} &= 105°30'00'' \\ R_b &= 26{,}243{,}052.74 \text{ ft} \\ Y_o &= 461{,}675.83 \\ \ell &= 0.6306895773 \end{aligned}$$

The applicable tabular values are

Latitude	R (ft)	Tabular Difference for $1''$ of Latitude (ft)
$39°51'$	25,508,572.22	101.18867
$52'$	25,502,500.90	101.18933
$53'$	25,496,429.54	101.19000
$54'$	25,490,358.14	101.19083
$55'$	25,484,286.69	101.19133

What is the latitude of station Thornton 2?

(A) $39°48'20.''99$
(B) $39°51'37.''41$
(C) $39°54'29.''78$
(D) $39°52'15.''59$

Solution: There is more than one method of finding the latitude of a station, given the state plane coordinates. One method is to calculate θ, and then find the value of R, the distance from the apex of the cone to the station. With this data, the latitude of the station can be derived from the given tabular values.

First, find X' from

$$\begin{aligned} X' &= X - C \\ &= 2{,}146{,}262.93 - 2{,}000{,}000 \\ &= 146{,}262.93 \end{aligned}$$

Next, find θ from

$$\begin{aligned} \tan\theta &= \frac{X'}{R_b - Y} \\ &= \frac{146{,}262.93}{26{,}243{,}052.74 - 742{,}549.22} \\ &= \frac{146{,}262.93}{25{,}500{,}503.52} \\ &= 0.005735688 \\ \theta &= +00°19'43.''06 \end{aligned}$$

Next, find the value of R from

$$\begin{aligned} R &= \frac{R_b - Y}{\cos\theta} \\ &= \frac{26{,}243{,}052.74 \text{ ft} - 742{,}549.22 \text{ ft}}{\cos +00^\circ 19' 43''.06} \\ &= \frac{25{,}500{,}503.52 \text{ ft}}{0.999983551} \\ &= 25{,}500{,}922.98 \text{ ft} \end{aligned}$$

Find the corresponding value of ϕ by interpolating from the tabular information.

Latitude	R (ft)	Tabular Difference for 1″ of Latitude (ft)
39°51′	25,508,572.22	101.18867
52′	25,502,500.90	101.18933
?	25,500,922.98	calculated value of R
53′	25,496,429.54	101.19000

Find the difference in R from that of the next smaller latitude.

$$\begin{array}{r} 25{,}502{,}500.90 \text{ ft} \\ \underline{-25{,}500{,}922.98 \text{ ft}} \\ 1577.92 \text{ ft} \end{array}$$

Divide the difference by the corresponding tabular difference for 1 sec of latitude.

$$\frac{1577.92 \text{ ft}}{101.18933} = 15''.594$$

Finally, add the seconds to the next smaller latitude to find the latitude of the station.

$$\begin{array}{rl} & 39^\circ 52' 00''.00 \\ & \underline{00^\circ 00' 15''.59} \\ \phi = & 39^\circ 52' 15''.59 \end{array}$$

Answer is (D)

72. The state plane coordinates of Thornton 2 in the Colorado Central zone are

$$\begin{aligned} X &= 2{,}146{,}262.93 \\ Y &= 742{,}549.22 \end{aligned}$$

The constants for the zone are

$$\begin{aligned} C &= 2{,}000{,}000 \text{ ft} \\ \text{central meridian} &= 105^\circ 30' 00'' \\ R_b &= 26{,}243{,}052.74 \text{ ft} \\ Y_o &= 461{,}675.83 \\ \ell &= 0.6306895773 \end{aligned}$$

The value of θ for the station is $+00^\circ 19' 43''.06$.

What is the longitude of Thornton 2?

(A) $104^\circ 58' 44''.18$
(B) $104^\circ 44' 21''.87$
(C) $104^\circ 32' 12''.09$
(D) $104^\circ 56' 55''.99$

Solution: When the mapping angle is given, the calculation of $\Delta\lambda$, or change in longitude, is fairly straightforward. The change in longitude can then be applied to the central meridian to find the longitude of the station.

Calculate $\Delta\lambda$ from

$$\begin{aligned} \Delta\lambda &= \frac{\theta}{\ell} \\ &= \frac{+00^\circ 19' 43''.06}{0.6306895773} \\ &= +00^\circ 31' 15''.820 \end{aligned}$$

Then apply the change in longitude to the central meridian.

$$\begin{aligned} \lambda &= \text{central meridian} - \Delta\lambda \\ &= 105^\circ 30' 00''.00 - (+00^\circ 31' 15''.820) \\ &= 104^\circ 58' 44''.18 \end{aligned}$$

$\boxed{\text{Answer is (A)}}$

73. The elevations and latitudes of Thornton 2 and 400 for the Colorado central zone are as follows.

Thornton 2:

$$\phi = 39°52'15''.594$$
$$\text{elev} = 5441 \text{ ft}$$

400:

$$\phi = 39°50'56''.090$$
$$\text{elev} = 5246 \text{ ft}$$

The applicable tabular information is

ϕ	R (ft)	Tabular Difference $1''$ ϕ (ft)	Scale as a Ratio
39°46′	25,538,928.21	101.18533	1.0000033
47′	25,532,857.09	101.18600	1.0000068
48′	25,526,785.93	101.18667	1.0000103
49′	25,520,714.73	101.18717	1.0000139
50′	25,514,643.50	101.18800	1.0000175
39°51′	25,508,572.22	101.18867	1.0000213
52′	25,502,500.90	101.18933	1.0000252
53′	25,496,429.54	101.19000	1.0000291
54′	25,490,358.14	101.19083	1.0000331
55′	25,484,286.69	101.19133	1.0000372

The distance measured between the two stations is 10,206.50 ft.

What is the grid distance between Thornton 2 and 400?

(A) 10,204.13 ft
(B) 10,203.89 ft
(C) 10,206.74 ft
(D) 10,204.92 ft

Solution: The sea-level factor will contribute the largest change to the measured distance. The traditional calculation of the factor uses the approximate radius of 20,906,000 ft for the reference spheroid.

$$\begin{aligned}\text{sea-level factor} &= \frac{20{,}906{,}000 \text{ ft}}{\text{average elevation of stations} + 20{,}906{,}000 \text{ ft}} \\ &= \frac{20{,}906{,}000 \text{ ft}}{5343.5 \text{ ft} + 20{,}906{,}000 \text{ ft}} \\ &= \frac{20{,}906{,}000.00 \text{ ft}}{20{,}911{,}343.50 \text{ ft}} \\ &= 0.99974447\end{aligned}$$

The scale factor is calculated for the stations at each end of the line and an average of the two taken. The scale factor is based on the latitude in the Lambert projection and found by interpolation.

scale factor for Thornton 2:

39°51′	25,508,572.22	101.18867	1.0000213
52′	25,502,500.90	101.18933	1.0000252
39°52′15.″594		.....	?
53′	25,496,429.54	101.19000	1.0000291

First, find the difference in scale factor for a full minute of latitude from 39°52′ to 39°53′.

$$\begin{array}{rr} 39°53' & 1.0000291 \\ 39°52' & \underline{-1.0000252} \\ & 0.0000039 \end{array}$$

Next, find the proportional change in the scale factor for 15.″594 over the same interval.

$$\begin{aligned}\frac{x}{15.''594} &= \frac{0.0000039}{60''} \\ 60''x &= (0.0000039)(15.''594) \\ x &= \frac{0.00006082}{60''} \\ &= 0.0000010\end{aligned}$$

This is the increase appropriate to the scale factor over 15.″594.

39°51′	25,508,572.22	101.18867	1.0000213
52′	25,502,500.90	101.18933	1.0000252
39°52′15.″594	.....	.....	1.0000262
53′	25,496,429.54	101.19000	1.0000291

Repeating the same process for point 400 yields a scale factor for that point of 1.0000210.

The average of the two scale factors at each end of the line is the scale factor for the line.

$$\frac{1.0000262 + 1.0000210}{2} = 1.0000236$$

The scale factor for the line is 1.0000236, and the sea-level factor is 0.99974447. The product of these two factors is the grid, or combination, factor for the line.

$$\begin{aligned}\text{grid factor} &= (\text{sea-level factor})(\text{scale factor})\\ &= (0.99974447)(1.0000236) = 0.99976806\end{aligned}$$

The product of the measured distance, 10,206.50 ft, and the grid factor, 0.99976806, is the grid distance.

$$(10{,}206.50 \text{ ft})(0.99976806) = 10{,}204.13 \text{ ft}$$

Answer is (A)

GOVERNMENT ADMINISTRATION OF STATE PLANE COORDINATE SYSTEMS

74. Which agency established the design of the state plane coordinate systems used in the U.S.?

(A) U.S. Geological Survey
(B) U.S. Coast and Geodetic Survey
(C) Bureau of Land Management
(D) Department of the Interior

Solution: The original request for a state plane coordinate system was submitted to the U.S. Coast and Geodetic Survey by an engineer from a state highway department in 1933. Within a year of the establishment of the North Carolina Coordinate System, a similar system had been designed for every state by Dr. O. S. Adams, a mathematician in the division of geodesy of the U.S. Coast and Geodetic Survey.

Answer is (B)

75. As the transition from NAD 27 to NAD 83 continues, what organization is ultimately responsible for legalization of the necessary changes in the various state plane coordinate systems?

(A) National Geodetic Survey
(B) U.S. Geological Survey
(C) Federal Board of Surveys and Maps
(D) legislature of each state

Solution: The individual state legislatures have jurisdiction over the legal status of the use of state plane coordinate systems.

Answer is (D)

76. Where are state plane coordinate systems available?

(A) Every state and possession of the U.S. has a state plane coordinate system.
(B) Thirty-seven states have state plane coordinate systems.
(C) Every state except Hawaii has a state plane coordinate system.
(D) Every public lands state has a state plane coordinate system.

Solution: Every state and possession of the U.S. has had a state plane coordinate system designed for it.

Answer is (A)

77. Which of the following organizations publishes geographical and state plane coordinates in both NAD 27 and NAD 83 for its national network of geodetic control stations?

(A) Bureau of Land Management
(B) Bureau of Reclamation
(C) Federal Geodetic Control Committee
(D) National Geodetic Survey

Solution: Formerly known as the U.S. Coast and Geodetic Survey, the National Geodetic Survey maintains and publishes the coordinates of geodetic control stations across the U.S.

Answer is (D)

78. Which of the following organizations publishes standards and specifications for the surveying of geodetic control networks?

(A) Federal Board of Surveys and Maps
(B) Bureau of Reclamation
(C) Federal Geodetic Control Committee
(D) U.S. Forest Service

Solution: The Federal Geodetic Control Committee was chartered on Dec. 11, 1968, and is responsible for coordinating the geodetic surveying activities of federal agencies. The organization publishes standards of geodetic surveying by various methods, including provisional standards for GPS surveying.

Answer is (C)